

4EB00 Special Topic: The Jet Engine

In the lectures a special topic is presented that deals with the intricacies of combustion within thermodynamics. The application is [The Jet Engine](#), as treated in

Section 8.2: Jet Engines
Section 9.4: Turbojet Engines

(First edition)
(Second Edition)

In Turns' book.

The Jet Engine falls into the category of so-called 'heat engines', which is a typology for all engines that convert heat into work (such as the steam engine and Stirling engine amongst others). In this case, the "heat" is provided by combustion, so that the jet engine falls into the "combustion engines" sub-class (just like the petrol and diesel engine).

Combustion is a key ingredient in this application; it drives it. That is why we will pay attention to how **combustion is contained (or perhaps even hidden) in the thermodynamic equations** for conservation of mass and energy. The following paragraphs from Turns' book are first discussed in the oral lectures:

Turns, 1st edition

Section 2.9: Ideal gas mixtures
Section 3.4: Reacting systems
Section 5.2b: Reacting systems

2nd edition

Chapter 10: Ideal gas mixtures
Chapter 12: Reacting Systems

Turns' book, however, mainly uses simple models for heat capacity, enthalpy and entropy to model all stages in the turbojet. In this part of 4EB00, we will make use of **a database and the associated MATLAB functions of component properties** (Nasa tables) to follow the same control volume approach (section 9.4c 2nd ed, or 5.2b 1st ed) but with a more advanced/consistent approach.

The database contains the coefficients for the temperature-dependent polynomials that describe various properties for each component as a function of the temperature (think of, for example, the heat capacity of N₂ as a function of the temperature, c_p , N₂ (T)). This approach is widely used in Academics and R&D in industries where combustion plays an important role (IC- and Jet engines, furnaces, power production etc..). For this assignment, **a small but relevant database and the associated functions (eg CpNasa) are made available via canvas** (you do not have to program the base functions nor generate the database of polynomial coefficients yourself).

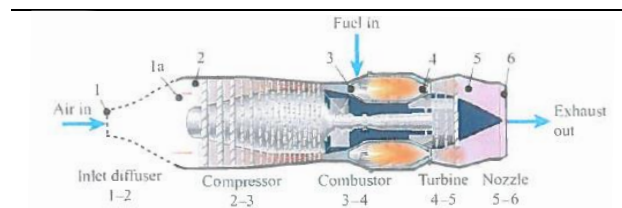


FIGURE 8.14

Figure 1:

Figure 8.14 Turns (1st ed)

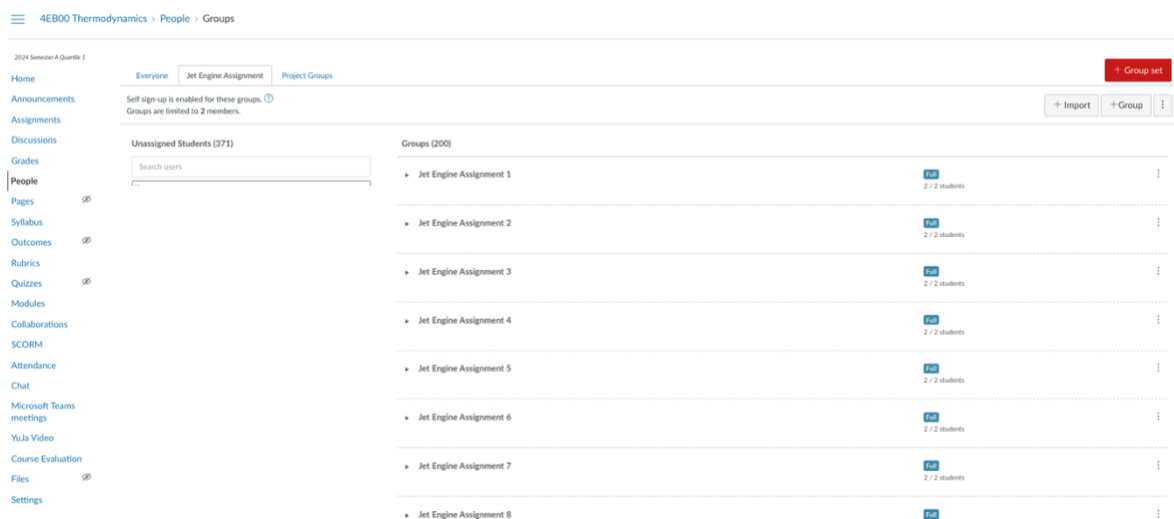
Figure 9.39 (2nd ed)

The set-up for this special assignment is as follows

1. **2hr oral lectures (1st and 2nd hr 23-9-2025)** in which the material is treated, including some simple exercises to get used to the database
2. **4hr guided selfstudy (3rd and 4th hr 23-9-2025 and 3rd and 4th 30-09-2025)**. These follow the same approach and locations as the general GS of the thermodynamics course.

The use of MATLAB is a requirement, and you need to be aware of the basic functionality of MATLAB, e.g., simple matrix, vector operations, and use of functions in particular.

The work is carried out in groups of 2. You must put together your own group. Groups (Jet Engine Assignment) are available on Canvas.



The result of the assignment is a working MATLAB script for the cycle analysis of a jet engine (see pages 551-554 of Turns 1st edition, 556-560 of 2nd edition) but applying the theory and approach treated in the lectures. The difference with the implementation of Turns is the use of the realistic component properties from the database and the use of conservation of energy and entropy. **Engineering approximations, like the Poisson relations, Lower Heating Value etc. should not be needed to perform the computations.** The assignment is finalized with a compact report, based on a template available on Canvas, in which you explain your model and present the results. **The submission (online) should be a single zip-file per group containing the MATLAB-model and a pdf of the report according to the following format**

GroepNr_report.pdf

GroepNr_scripts.zip (containing a working script when unzipped)

A word-template for the report is available on Canvas and should be strictly followed.

The **deadline for the submission is 7th of October** and available till **14th of October**. **Late** submissions will be graded with an 8 maximally according to **Grade = min(Grade,8)**. Assignment opens on Monday 22 September.

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